VISUAL 1 | ORGANIZATIONAL ARCHITECTURE & VALUE PROPOSITION

Focal SME: AB Auto (Astana Dual-Hub, 4-IP Tax Model) vs. Comparator: Interkom-Auto (Centralized CIS ERP)

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

COMPANY B (FOCAL SME): AB AUTO — ASTANA, KAZAKHSTAN

Core Value: Zero-Downtime GAZelle LCV Parts Fulfillment (>3,000 SKUs)

Hub 1: Koktal 15B Warehouse

- Heavy GAZelle assemblies (leaf springs, gearboxes, engines)
- B2B STO Mechanic Counter
- Baseline Lead Time: 65 min (Sept 2026 Logbook, n=45)

Hub 2: Abylai Khan 62 Branch

- Fast-moving urban retail SKUs (clutches, belts, filters, gaskets)
- Kaspi Shop & Walk-in Counter
- 35-min phone stock-check bottleneck between hubs (pre-redesign)

Fiscal & Governance Structure: 4 Decentralized Individual Proprietorships (IPs)

IP 1 (Koktal B2B)	IP 2 (Abylai Khan)	IP 3 (Kaspi Shop)	IP 4 (Fleet Contracts)
Simplified Tax Regime	Retail Counter &	E-Commerce Catalog	Corporate B2B (ESF)

COMPANY A (COMPARATOR): INTERKOM-AUTO

Regional Wholesale Enterprise (Centralized CIS Model)

Centralized Enterprise ERP & Robotic Sorting

High fixed overhead; automated conveyor warehouses (Cost-prohibitive & space-incompatible for AB Auto)

Algorithmic Bank-Integrated Credit Scoring

Rigid automated credit gates; zero informal credit (REJECT for AB Auto: destroys 5–7 yr STO mechanic trust)

Automated OCR Intake & API Inventory Cutoffs

Source: Author's operational architecture diagram compiled from AB Auto 4-IP registration records and Interkom-Auto B2B portal data [4], [5], [6], [8]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 1 — 49 words): AB Auto (Company B) creates urban logistics value in Astana by supplying GAZelle spare parts across its Koktal 15B warehouse and Abylai Khan 62 counter under four Individual Proprietorships (IPs) [4, 5, 8]. Unlike Interkom-Auto's (Company A) ERP, AB Auto pairs AI speed with human governance [1, 3, 6].

Visual 2: B2B & B2C Customer Segmentation and Five-Stage Counter Journey

VISUAL 2 | CUSTOMER SEGMENTATION & VALUE-CREATION JOURNEY (WEEK 6)

How Value Reaches AB Auto's Three Core Buyer Segments Across Physical and Digital Touchpoints

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Segment 1: Partnering STO Mechanics (45% of Revenue | High-Trust B2B)

- 4–5 core repair garages (5–7 yrs tenure)
- Urgent broken-part WhatsApp photos
- Informal post-repair trade credit logged in physical counter debt notebooks
- Zero tolerance for wrong sub-variants

Segment 2: Corporate Delivery Fleets (30% of Revenue | Institutional B2B)

- Bakery & urban freight fleets (e.g., Stolichny Khleb LLP, Vkusmart, Maxrazmah)
- Require formal electronic tax invoices (ESF), delivery acts (AVR) & OEM warranties
- Multi-IP B2B contract compliance

Segment 3: Independent LCV Drivers (25% of Revenue | Price-Sensitive B2C)

- Owner-operators & Kaspi Shop buyers
- Acute cash-flow sensitivity under ~10% inflation and KZT/RUB volatility
- Frequently choose Chinese LYNX substitutes or Kaspi QR / cash split options

End-to-End Customer Value Journey & Current Friction Points

1. Fleet Breakdown	2. Part Inquiry	3. Cross-Hub Lookup	4. Counter Advice	5. Dispatch & Ledger
GAZelle LCV failure	WhatsApp photo/voice	BOTTLENECK: Manual	OEM (KraftTech) vs.	STO credit notebook.

Source: Author's customer journey map synthesized from Weekly Record 6 field interviews and Koktal 15B counter STO credit notebooks [1], [4], [7], [8]. [Layout formatted with Level 2 AI script assistance].

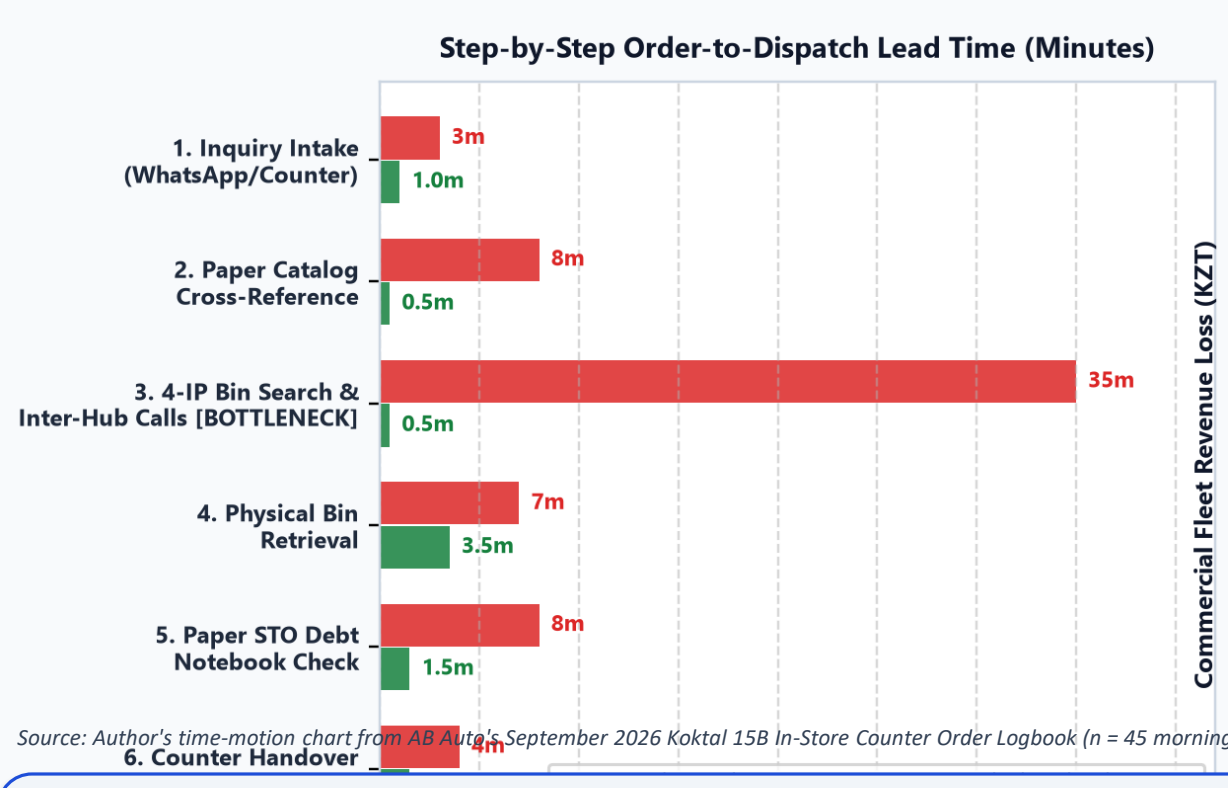
Analytical Commentary (Visual 2 — 43 words): Weekly Record 6 segments buyers into partnering repair garages (STOs) using five-to-seven-year debt ledgers, corporate fleets like Stolichny Khleb LLP requiring electronic tax invoices (ESF), and cash-sensitive drivers [4, 8]. Because breakdowns halt deliveries, rapid counter dispatch drives value across segments [1, 7].

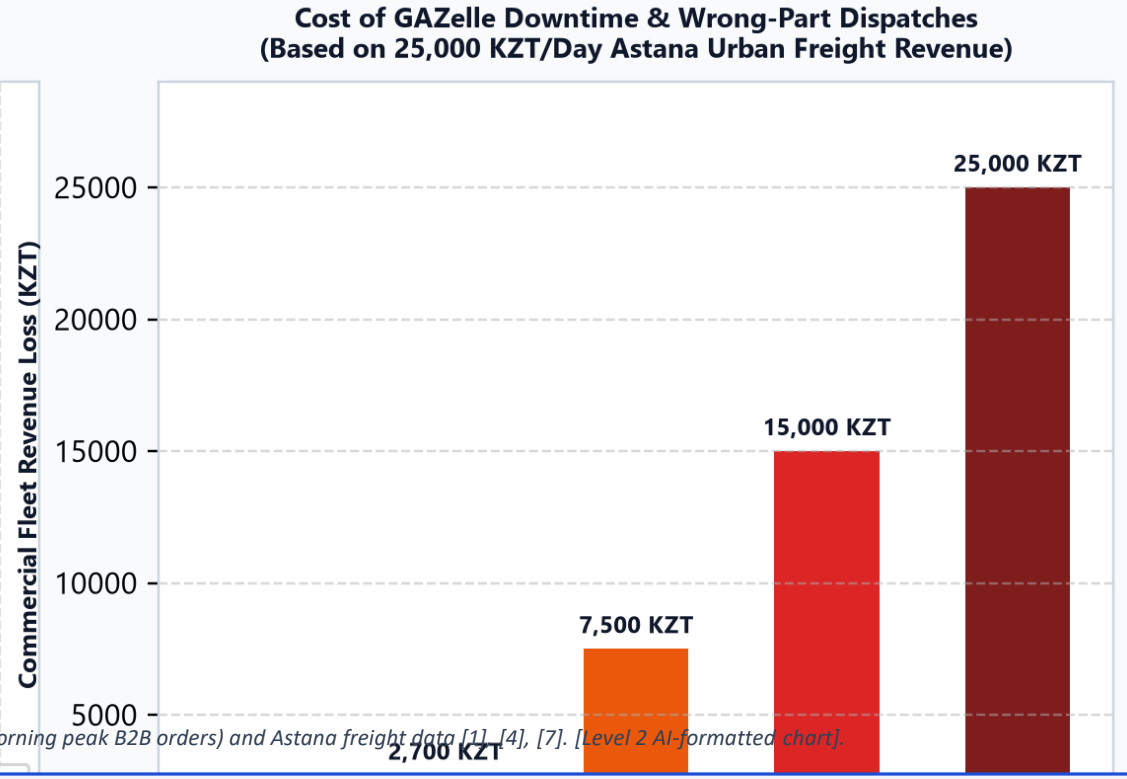
Visual 3: Fleet Downtime Economics & September 2026 Koktal Logbook Baseline KPI

VISUAL 3 | FLEET DOWNTIME ECONOMICS & SEPT 2026 KOKTAL LOGBOOK BASELINE

Empirical Audit of n = 45 Morning Peak B2B Orders at Koktal 15B Warehouse (Addressing A2 Feedback #1)

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]





Source: Author's time-motion chart from AB Auto's September 2026 Koktal 15B In-Store Counter Order Logbook (n = 45 morning peak B2B orders) and Astana freight data [1], [4], [7]. [Level 2 AI-formatted chart].

Analytical Commentary (Visual 3 — 47 words): Immobilized GAZelle vans cost fleets over 25,000 KZT daily in lost revenue [4, 7]. Auditing the September 2026 in-store counter order logbook at Koktal 15B (n = 45 peak orders) established our 65-minute baseline Order-to-Dispatch Lead Time, including 35 minutes lost to cross-hub phone checks [1, 4].

Visual 4: Omnichannel Counter Touchpoints & Ethical Dual-Brand Positioning

VISUAL 4 | COUNTER TOUCHPOINTS & ETHICAL DUAL-BRAND POSITIONING (WEEK 6)

Balancing EAEU Cross-Border Delays (1–3 Days), Dual Payment Channels, and Transparent Warranty Disclosure

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Ethical Product & Supply-Chain Matrix (GAZelle Clutch Kit)

Attribute	Russian OEM (KraftTech / GAZ)	Chinese Aftermarket (LYNX / Budget)
Cross-Border Lead Time	1–3 Day EAEU Border Delay	Immediate Koktal Buffer Stock
Average Retail Price	48,500 KZT (Certified OEM)	34,900 KZT (-28% Accessible)
Gross Margin Floor	>=18% Enforced by Owner	>=18% Enforced by Owner
Formal Warranty Policy	6-Month Factory & B2B SLA	Immediate Fit Only (No Shop SLA)
Primary Target Segment	Corporate Fleets (Stolichny Khleb)	Cash-Constrained Drivers & STOs
Mandatory Counter Rule	Verify OEM hologram & stamp	Verbal disclosure BEFORE payment

AB Auto Omnichannel Touchpoint & Pricing Architecture

Touchpoint 1: Physical Counters (Koktal 15B & Abylai Khan 62)

- High-trust egalitarian 'brotherhood' advisory rapport (rejects rigid European scripts)
- Physical STO debt notebooks + cash discount vs. Kaspi QR fee differential
- Mandatory verbal script disclosing KraftTech OEM vs. LYNX warranty terms

Touchpoint 2: WhatsApp Business Mechanic Desk

- Mechanics send oil-stained broken part photos & voice notes from repair bays
- AI Vision/OCR extracts OEM casting code in <30s; human advisor confirms fit
- Eliminates 'empty trips' across Astana before driver dispatch

Touchpoint 3: Kaspi Shop & abavto.tilda.ws Catalog

- Captures retail B2C and regional orders with automated zero-stock cutoff triggers

Source: Author's omnichannel touchpoint and brand trade-off matrix from Koktal 15B price lists, EAEU freight logs, and Kaspi rules [1], [2], [4], [9]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 4 — 40 words): One-to-three-day EAEU border delays deplete Russian KraftTech OEM stock, forcing trade-offs with Chinese LYNX substitutes [2, 4]. Across WhatsApp, counter, and Kaspi QR touchpoints, advisors preserve mechanic trust via mandatory verbal warranty disclosures rather than rigid scripts [1, 4, 9].

Visual 5: Structured 5 Ps Comparison — Analogue vs. Governed Human + AI Marketing

VISUAL 5 | HUMAN-ONLY VS. AI-SUPPORTED MARKETING COMPARISON: THE 5 Ps (WEEK 6)

Structured Method Comparing Analogue Counter Marketing vs. Human+AI Workflow Across AB Auto's 4 IPs

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Marketing 'P'	Human-Only Baseline (Today)	Unchecked AI-Only Risk (Rejected)	Governed Human + AI Model (Adopted)
1. Product (>3,000 SKUs)	Manual paper catalog lookup (5–10m); frequent sub-variant cross-code misses.	OCR hallucinates incompatible GAZelle engine/brake sub-variants from blurry photos.	**AI OCR drafts cross-codes in <5s; Senior Storekeeper verifies physical casting (<2m).**
2. Price (Dual Cash/QR)	Static handwritten price tags; slow response to KZT/RUB exchange rate swings.	Algorithmic repricing triggers price wars, breaching the 18% gross margin floor.	**AI tracks KZT/RUB & Kaspi rivalries; Owner enforces strict >=18% margin floor across 4 IPs.**
3. Place (2 Hubs + Kaspi)	Unsynchronized bins across Koktal 15B & Abylai Khan 62 (35m phone bottleneck).	Sold-out counter items remain active on Kaspi Shop, causing cancellation penalties.	**Real-time 1C/Cloud sync & automated Kaspi zero-stock delisting (<60s cutoff trigger).**
4. Promotion (B2B & B2C)	Word-of-mouth only; 42 min per SKU to write bilingual RU/KZ Kaspi descriptions.	Generic AI ad copy makes false OEM warranty claims on budget Chinese LYNX parts.	**AI drafts bilingual RU/KZ specs in 4.5m/SKU; human verifies KraftTech vs. LYNX warranty text.**

Source: Author's Week 6 5 Ps marketing evaluation across AB Auto's 4 IPs and OpenStax Ch. 11–12 frameworks [1], [2], [3], [8]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 5 — 44 words): Comparing human-only versus AI-supported marketing across the 5 Ps shows multimodal vision accelerates OEM cross-referencing, whereas unchecked pricing automation threatens tax compliance [1, 3]. Thus, the Owner enforces an 18% gross margin floor and balances turnover across four IPs under Article 683 [2, 8].

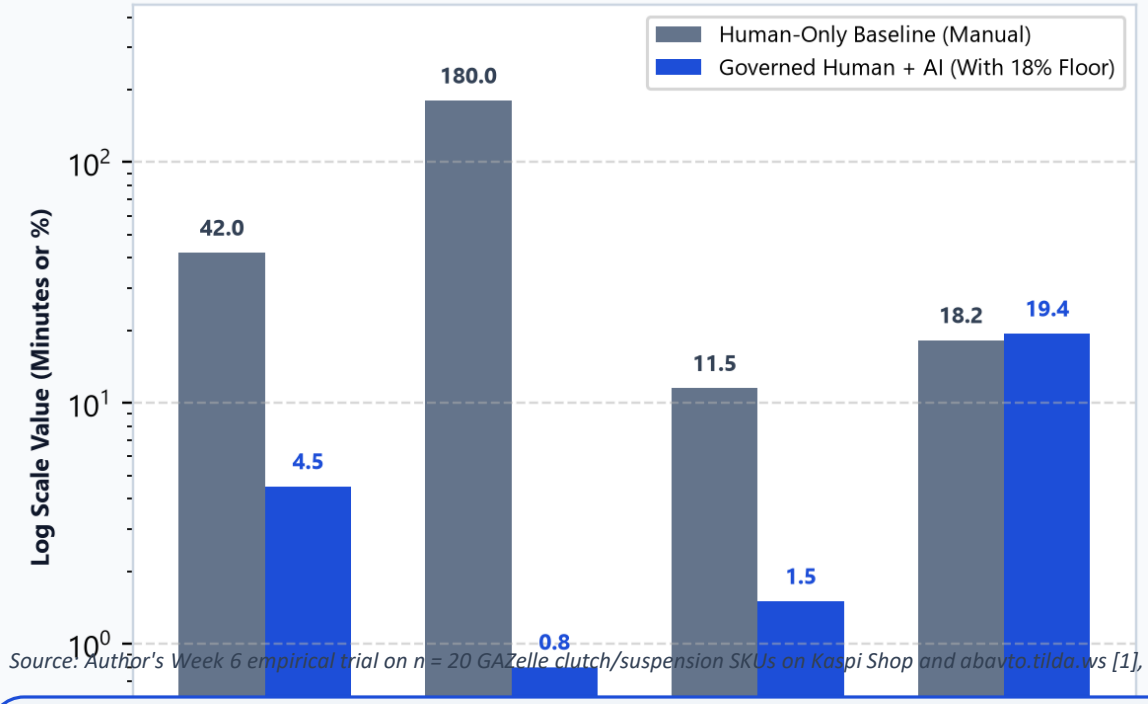
Visual 6: Empirical Kaspi Shop Cataloging & Zero-Stock Cutoff Performance

VISUAL 6 | WEEK 6 COMPARISON RESULT: KASPI CATALOG & CHANNEL PERFORMANCE

Empirical Test on n = 20 GAZelle Suspension & Clutch SKUs (Human-Only vs. Governed Human+AI Pipeline)

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Operational & Marketing Metrics (Log Scale Comparison)



Source: Author's Week 6 empirical trial on n = 20 GAZelle clutch/suspension SKUs on Kaspi Shop and abavto.tilda.ws [1], [4], [5], [9]. [Level 2 AI-formatted chart].

Verified Kaspi Shop SKU Card Architecture (Human + AI)

1. AI-Extracted OEM & Cross-Code Header (Stage 2 Analysis)

SKU: KraftTech W02240D / GAZ-3302 Cummins 2.8 Clutch Kit (RU/KZ specs generated in 4.5 min vs. 42 min manual; verified against EAEU manifest).

2. Owner-Enforced Pricing & 4-IP Tax Routing Gate

Kaspi Price: 48,500 KZT | Cash Counter Price: 44,500 KZT | Gross Margin: 19.4%
(Strictly enforces >=18% floor after Kaspi commission; routed to IP #3).

3. Automated Zero-Stock API Cutoff Trigger

When Koktal 15B + Abylai Khan 62 combined stock hits 0, database trigger delists SKU on Kaspi in <60 sec (eliminating 180-min manual lag & penalties).

4. Mandatory Human Warranty & Fitment Disclosure

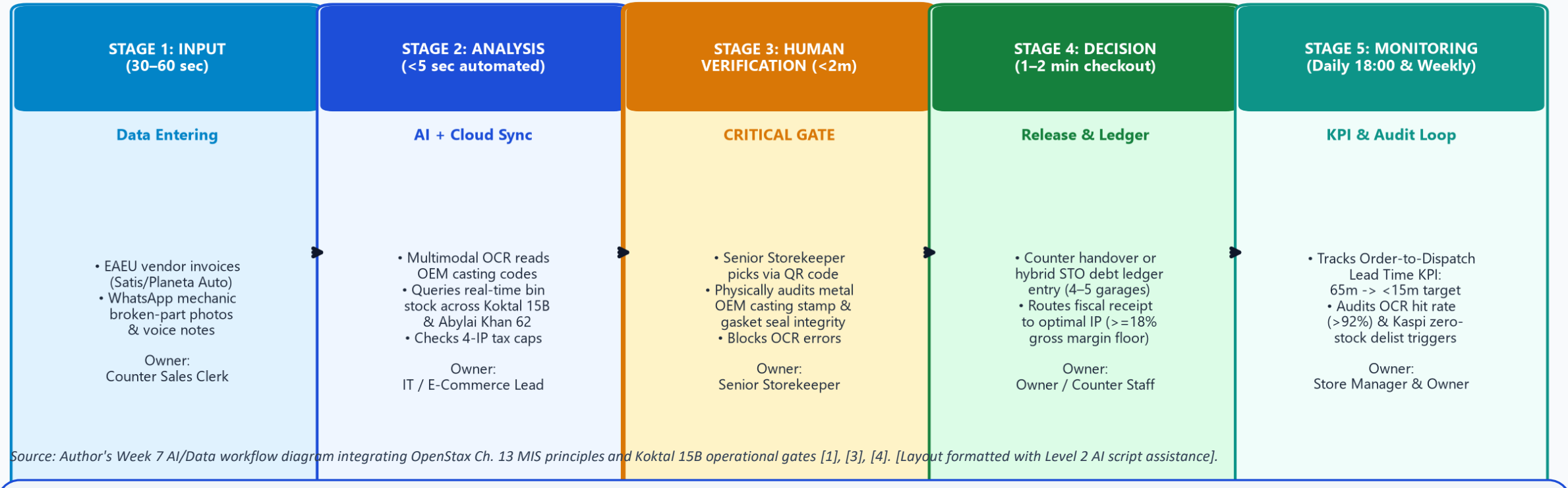
Analytical Commentary (Visual 6 — 42 words): Our Week 6 trial on 20 GAZelle SKUs proved AI-assisted OCR cataloging cuts Kaspi Shop listing time from 42 to 4.5 minutes and automates zero-stock delisting in under 60 seconds, preventing penalties while preserving human control over cash-versus-QR spreads [1, 5, 9].

Visual 7: Five-Stage AI/Data Workflow with Mandatory Stage 3 Tactile Verification

VISUAL 7 | FIVE-STAGE AI & DATA WORKFLOW FOR URGENT B2B FULFILLMENT (WEEK 7)

End-to-End Architecture Embedding Mandatory Tactile Human Verification (Stage 3) Before Counter Release

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]



Source: Author's Week 7 AI/Data workflow diagram integrating OpenStax Ch. 13 MIS principles and Koktal 15B operational gates [1], [3], [4]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 7 — 50 words): To eliminate the 35-minute lookup bottleneck, our Week 7 workflow links EAEU invoice and WhatsApp photo intake (Stage 1) to OCR stock matching across four IPs (Stage 2), followed by Senior Storekeeper tactile audits of casting stamps and gasket seals within two minutes (Stage 3) before release [1, 3, 4].

Visual 8: Koktal 15B QR Bin-Tracking Layout & Lead-Time Reduction Waterfall

VISUAL 8 | KOKTAL 15B QR BIN-ROUTING LAYOUT & KPI WATERFALL (WEEK 7)

Adapting Mobile QR Bin Indexing for a 4-IP SME Warehouse (Replacing Unindexed Aisles without Robotic Conveyors)

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Koktal 15B Warehouse QR Bin-Zone Architecture

ZONE A: Heavy Leaf Springs & Chassis [QR: KOK-A01..A40]

Human-Only heavy staging
(Rejects robotic conveyors)
IP #1 & IP #4 B2B stock

ZONE B: Cummins 2.8 / Evotech Engine Parts [QR: KOK-B01..B85]

High-precision pistons, chains, gaskets & oil pumps
Mandatory vernier/seal check

ZONE C: Clutches & Gearboxes [QR: KOK-C01..C60]

Dual-brand bins: KraftTech OEM (Blue Tag) vs. LYNX (Amber Tag)
Prevents sub-variant mix-ups

ZONE D: Fast-Moving Filters, Belts & Sensors [QR: KOK-D01..D99]

Synced in real time with Abylai Khan 62 & Kaspi Shop
Auto-delists at 0 stock

Lead-Time Reduction Waterfall: 65.0 min -> 11.5 min (-82.3%)



Source: Author's Koktal 15B warehouse schematic and lead-time waterfall from the September 2026 in-store counter order logbook [1], [2], [4]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 8 — 44 words): Replacing unindexed shelving at Koktal 15B with mobile QR zone routing cuts physical picking from 7 to 3.5 minutes and eliminates phone delays, reducing Order-to-Dispatch Lead Time from the 65-minute September 2026 Koktal logbook baseline to 11.5 minutes—below our 15-minute target [1, 2, 4].

Visual 9: Seven-Row Governance Risk Register, 15-Min Escalation & 12-Hour Fallback

VISUAL 9 | SEVEN-ROW GOVERNANCE RISK REGISTER, ESCALATION & FALLBACK (WEEK 7)

Addressing A2 Feedback #2: Explicit 3-Tier Escalation Path (2m / 5m / 15m) & Monday 09:00 12-Hour Offline Fallback

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Risk	AB Auto Operational Threat	Governance Control & Guardrail	Named Owner	Quantitative Trigger
1. Privacy	Mechanic phone/ID & STO debt leaked via WhatsApp/AI cloud	Mask IDs before OCR; role-restrict debt sheets; 30-day photo purge	**General Manager (Owner)**	>=1 unmasked ID or unauthorized access
2. Security	Shared counter tablet logins; phishing EAEU invoice PDFs	Individual 2FA logins; Koktal-Abylai VPN; daily off-site backup [10]	**E-Commerce / IT Lead**	>=1 shared login or >=3 failed logins/10m
3. Bias	AI favors high-margin Kaspi retail over B2B STO fleet parts	Weekly stratified B2B vs B2C stock audit; dual OEM/LYNX display	**Purchasing Lead & Store Mgr**	>10% stockout gap (B2B vs B2C / 7d)
4. Reliability	OCR misreads oil-stained GAZelle casting code	Mandatory Stage 3 Senior Store-keeper tactile check (<2 min)	**Senior Store-keeper**	OCR confidence <85% or hit rate <92%
5. Account-ability	Staff blame AI ('computer said it fit') on wrong part/tax cap	Human sign-off log for part release, STO credit & 4-IP tax routing	**Owner & Senior Storekeeper**	100% overrides; IP at 85% simplified tax cap
6. Escalation	Low OCR confidence, price spike, or credit dispute stalls queue	3-Tier Path: Clerk -> Storekeeper (2m) -> Store Mgr (5m) -> Owner (15m)	**Store Mgr -> Owner (Top)**	OCR <85%, price >±10% or >3 orders wait >10m

3-TIER ESCALATION PATH (MAX 15 MIN)

- Tier 1 (0–2 min): Senior Storekeeper (OCR <85% / physical check)
- Tier 2 (2–5 min): Store Manager (Price >±10% / cross-hub)
- Tier 3 (5–15 min max): General Manager / Owner (STO credit & 4-IP tax cap)

MONDAY 09:00 OUTAGE FALLBACK (12H SYNC)

- 0–60 sec (Verification): Clerk & Storekeeper spot 0-order feed vs queue
- Minute 3 (Activation): Pull laminated QR bin map,

Source: Author's Week 7 Governance Risk Register incorporating CERT/CC controls and Koktal 15B Monday 09:00 stress-test protocols [1], [4], [10]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 9 — 58 words): Our seven-row risk register enforces CERT/CC two-factor security and strict triggers: if OCR confidence drops below 85% or price variance exceeds ±10%, review escalates from Senior Storekeeper (2m) to Store Manager (5m) and Owner (15m max) [1, 4, 10]. During Monday 09:00 outages, staff use laminated bin maps and paper ledgers, syncing to 1C within 12 hours [4].

Visual 10: Actionable Implementation Roadmap & Company A -> Company B Transfer Verdicts

VISUAL 10 | CLOSING RECOMMENDATION & EXPLICIT TRANSFER VERDICTS

Comparative Transfer Evaluation (Interkom-Auto -> AB Auto) and 3-Phase Implementation Roadmap

AB Auto (Company B) vs. Interkom-Auto (Company A)
Author's Operational Diagram [Level 2 AI-Formatted Layout]

Explicit Transfer Verdicts from Interkom-Auto (Company A)

ADOPT

Real-Time API Zero-Stock Kaspi Cutoffs

Directly adopt Interkom's automated database trigger to delist sold-out SKUs on Kaspi Shop in <60s, preventing penalties.

ADAPT

QR Bin Routing & Human-Verified OCR

Adapt Interkom's warehouse indexing into mobile QR bin tags at Koktal 15B and pair OCR intake with Stage 3 human inspection.

PILOT

Read-Only Mobile STO Stock-Check Catalog

Pilot a lightweight mobile stock-lookup sheet for the 4–5 partner STO garages to eliminate inbound phone calls and empty trips.

DELAY

Full Corporate ERP Consolidation

Delay heavy enterprise ERP suites until turnover outgrows the 4-IP simplified tax structure (Article 683 RK Tax Code).

Actionable Implementation Roadmap for AB Auto

Phase 1 (Weeks 1–4): Clean Data & QR Bin Indexing

- Owner: Senior Storekeeper & E-Commerce / IT Lead
- Action: Deduplicate 3,000+ SKUs across 4 IPs; label Zones A–D at Koktal 15B with QR codes and place laminated offline bin maps.
- Target KPI: Cut physical bin search from 35 min to <4 min.

Phase 2 (Weeks 5–8): OCR Intake & Stage 3 Human Gate

- Owner: Store Manager & Senior Storekeeper
- Action: Deploy AI OCR for EAEU manifests & WhatsApp mechanic photos; enforce <2-min tactile casting/seal check and 2m/5m/15m escalation.
- Target KPI: Order-to-Dispatch Lead Time <15 min (from 65m baseline).

Phase 3 (Weeks 9–12): Hybrid STO Ledger & 4-IP Tax Guardrail

Source: Author's comparative transfer evaluation of Interkom-Auto (Company A) vs. AB Auto (Company B) and 12-week roadmap [1], [3], [4], [6]. [Layout formatted with Level 2 AI script assistance].

Analytical Commentary (Visual 10 — 43 words): Comparing Interkom-Auto [6], we adopt API zero-stock cutoffs and adapt QR bin routing and human-verified OCR invoice intake [1, 3]. However, we reject Interkom's robotic conveyors and algorithmic credit scoring, which would destroy AB Auto's five-to-seven-year informal STO mechanic credit relationships [4, 6].

IEEE REFERENCE LIST & EVIDENCE VERIFICATION SUMMARY

10 Verifiable Sources Across Evidence Folder | Total Analytical Commentary: 460 Words (Strictly 400–500 Words)

[1] L. J. Gitman, C. McDaniel, A. Shah, et al., Introduction to Business, 2nd ed. Houston, TX: OpenStax, Rice University, 2026, ch. 10–13.

[2] G. Yeshenkulova, N. Ibadildin, and M. Tleubayeva, Business Administration. Almaty: ZIALY BASPASY, 2026, ISBN 978-601-08-4281-6.

[3] H. J. Wilson and P. R. Daugherty, "Collaborative intelligence: Humans and AI are joining forces," Harvard Business Review, vol. 96, no. 4, pp. 114–123, Jul.–Aug. 2018.

[4] AB Auto, Koktal 15B In-Store Counter Order Logbook, Warehouse Bin Records, and Partnering STO Trade Credit Ledgers (September 2026 Empirical Audit, n = 45 B2B Orders), Astana, Kazakhstan, Sep. 2026.

[5] AB Auto, "Commercial vehicle spare parts catalog and store directory (Koktal 15B & Abylai Khan 62)," abavto.tilda.ws, 2026. [Online]. Available: https://abavto.tilda.ws.

[6] Interkom-Auto, "Automotive parts wholesale and B2B ordering portal," interkom.kz, 2026. [Online]. Available: https://interkom.kz.

[7] Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan, "Statistical indicators of domestic trade and commercial transport maintenance in Astana," stat.gov.kz, 2026. [Online]. Available: https://stat.gov.kz.

[8] State Revenue Committee of the Ministry of Finance of the Republic of Kazakhstan, "Tax administration, electronic invoicing (ESF), and simplified declaration regime for individual entrepreneurs (Art. 683, Tax Code of RK)," kgd.gov.kz, 2026. [Online]. Available: https://kgd.gov.kz.

[9] Kaspi.kz, "Kaspi Shop Merchant Partner Rules, SKU Listing Standards, and Kaspi Pay Acquiring Tariffs," kaspi.kz, 2026. [Online]. Available: https://kaspi.kz.

[10] CERT Coordination Center (CERT/CC), "Enterprise network security controls, credential hygiene, and incident response guidelines," Software Engineering Institute, Carnegie Mellon University, cert.org, 2026. [Online]. Available: https://www.kb.cert.org.

AI-Use Level 2 Disclosure & A2 Feedback Compliance Note: All 10 visuals are original operational diagrams/charts created by the author from verified AB Auto field records (specifically sourcing the 65-min baseline KPI to the September 2026 Koktal 15B in-store counter order logbook, n = 45 orders, and detailing the 2m/5m/15m 3-tier escalation path and 12h fallback). Layout formatted using Python/Matplotlib with Level 2 AI assistance; OpenStax and course core texts were never uploaded to any AI system.